



CHAPTER FIVE

SPEED AND AGILITY TRAINING

There are many different methods of training used to train athletes for improvement in sport performance. Each of these methods serves a purpose and can target certain physiological characteristics. It is very important for strength training and conditioning coaches to understand that these different methods of training all have different purposes. As described previously, when designing and implementing strength and conditioning programs, strength training and conditioning coaches need to have an objective for the training session and training cycle. This will ensure that programs target the physiological characteristics they are designed to target. The special methods of training reviewed in this chapter will allow for a greater understanding of different methods that can be implemented within strength training and conditioning programs to illicit improvements in performance through speed and agility training.

Introducing Plyometrics

A popular type of training exercise is plyometric training (2,3,19,20,32). This type of training uses jumping exercises to involve the neuromuscular system in rapid force development and improved use of tissue elasticity to improve power, rate of force development, and the ability to absorb force. A simple jumping exercise can be classified as a plyometric exercise if it involves an eccentric muscle action (lengthening of the muscle tissue), amortization phase (phase between eccentric and concentric muscle actions), and a concentric muscle action (shortening of the muscle tissue). These three components allow athletes to utilize the musculoskeletal system to produce greater force following the eccentric muscle action. For example, a countermovement jump involves a quick lowering of the body to produce eccentric tension in the relevant musculature which is then followed by a transition from down to up, the amortization phase, and then the athlete develops concentric or shortening tension in the relevant muscles as he/she performs the “up” phase of the jump.

Plyometric training is very stressful, but the stress comes from the mechanical impact demands placed on relevant tissues and the ability of the nervous system to respond rapidly to these quickly applied forces. It is imperative that the musculoskeletal system is given time to adapt to the demands of a plyometric training program (6,7,8). Plyometric exercise sessions should begin with

a general warm-up, dynamic stretching, and specific warm-up movements. The specific warm-up for plyometric training should consist of low-intensity, dynamic movements. During the warm-up, the athlete gradually progresses to actual plyometric or rebound types of exercises.

Plyometric exercise is not inherently dangerous; however, developing an adequate base of strength and power should be seen as a prerequisite for serious plyometric training. Low-intensity plyometrics, such as running, leaping, hopping, and so forth, can be used with novice athletes. However, drop jumps, shock training (dropping from heights greater than 1 m, and then landing and jumping again) should be reserved for advanced athletes. Landing surfaces, proper footwear, number of repetitions, base strength, and speed of execution all interact in plyometric training and affect the intensity, volume, and/or injury potential of the exercise.

Plyometrics are merely a part of an overall strength training and conditioning program that includes strength, speed, agility, aerobic and anaerobic training, flexibility training, and proper nutrition. Plyometrics have a purpose, which is usually described as power and speed training, but plyometrics are not a panacea (9,13,21,24,28).

Plyometrics

Virtually all of the things required during athletic events involve one’s ability to exert force in a rapid manner. This ability to exert force can be in the form of jumping, hopping, and changes of direction (1). Improving one’s ability to exert force in a rapid manner can have a positive influence on sports performance. Prior to improving one’s ability to exert force rapidly, it is important that athletes can absorb force during various movements. Improving absorption of force involves an adequate strength base, and strong and elastic connective tissue.

The Stretch-Shortening Cycle

A common mistake is to confuse plyometrics (the exercise the athlete performs) with the stretch-shortening cycle (the mechanism underlying the tissue response). The stretch-shortening cycle involves the tissue response to the three phases

Phase	Action	Physiological Event
I—Eccentric	Stretch of the agonist muscle	- Elastic energy is stored in the series elastic component - Muscle spindles are stimulated
II—Amortization	Pause between phases I and III	- Type Ia afferent nerves synapse with alpha motor neurons. - Alpha moto neurons transmit signals to agonist muscle group.
III—Concentric	Shortening of agonist muscle fibers	- Elastic energy is released from the series elastic component - Alpha motor neurons stimulate the agonist muscle group

Figure 5-1. Stretch-Shortening Cycle Phases

of a plyometric exercise as described above. Figure 5-1 shows the three phases of the stretch-shortening cycle along with the underlying physiology that accompanies each phase.

Deceleration and Jump Training for Novice Athletes

Conventional resistance training should be used prior to plyometrics in order to promote general strength gains and connective tissue development (10,11). One of the other methods used to promote general strength and connective tissue development is deceleration/jump training. This type of training does not involve high levels of orthopedic demands, and thus reduces the likelihood of injury. These exercises involve an emphasis on landing mechanics. Putting a large emphasis on landing mechanics also allows for the development of eccentric strength in a dynamic environment. The key to this type of training is to promote dynamic eccentric strength and the landing mechanics of each exercise. Coaches can use jumps in place (drills that involve jumping and landing in the same spot), multiple hops, and jumps that cover distance. For all of these types of jumps, emphasis should be on the landing portion of the jump, and one's ability to decelerate and avoid reaching the extreme range of motion limit of the joints. One of the major distinctions between plyometric training and traditional strength training and conditioning is that plyometric training targets the nervous system more than the muscular system. The nervous system is trained to respond rapidly and without hesitation. Factors that influence the intensity of plyometric exercises are shown in Figure 5-2.

Speed and Agility

Two methods of training that have been utilized for many years are speed and agility training. Speed and agility training usually involves moving at high rates of speed. Speed training involves running and other rapid movements (4,15,18,22,29). There are different types of speed:

Reactive Speed: The ability to react to a stimulus in the shortest possible time (reactive speed involves both simple and choice reactions)

Active Speed: The ability to move quickly and precisely during short duration activities of less than 6 s (active speed involves noncyclic movements)

Frequency Speed: The ability to run, walk, row, climb, swim, and so forth involving cyclic movement activities

Complex Speed: The ability to holistically combine reactivity, perception, spatial relationships, and so forth to produce movements that are coordinated and goal directed, and involves primarily noncyclic activities (29)

Agility training combines moving at high speeds with rapid changes of direction. Agility training is commonly used to mimic competitive situations and improve the ability of athletes to simulate the movement and perceptual demands of the sport. This method of training usually involves movement patterns that target multiple changes of direction over the course of an activity. In many cases, speed training and agility training are performed together (i.e., within the same training session or within the same training cycle). Speed and agility methods of training involve high-intensity work and it is imperative that strength training and conditioning coaches monitor rest periods and volume to decrease the chance of injury during the course of a training session. Utilizing speed and agility training can improve the transfer of performance from the accumulation effect of the training sessions to competitive situations (4,5).

Linear Speed

One of the key elements of a conditioning program is ensuring the athletes move as efficiently and effectively as possible. In relation to speed, it is essential that an effective sprinting technique be stressed from day one. The length of each step (step length, SL) and the number of steps taken over a specific distance (step frequency, SF) determine the speed or velocity of an athlete (velocity = SL x SF) (16).

In most sports, the key factor in linear speed is not the top speed achieved, but the ability to accelerate (i.e., the first 10 yards) (25,30,31). Acceleration refers to the rate of change in velocity over time. Therefore, athletes need to be able to reach maximum velocity in as little time as possible. During the first 10 yards from

Factor	Effect
Points of contact	The ground reaction force during single-leg lower body plyometric places more stress on an extremity's muscles, connective tissues, and joints than during double-leg plyometric drills
Speed	Greater speed increases the intensity of the drill
Height of the drill	The higher the body's center of gravity, the greater the force on landing
Bodyweight	The greater the athlete's bodyweight, the more stress is placed on the muscles, connective tissues, and joints. External weight (in the form of weight vests, ankle weights, and wrist weights) can be added to the body to increase a drill's intensity

Figure 5-2. Plyometric Intensity Factors

the start, step length will be relatively small with a high step frequency (3). As force can only be applied while the foot is in contact with the ground, the more time the foot is on the ground, the greater amount of force can be produced through each step. As the athlete reaches maximum velocity (by 30 – 40 yards), step length will likely increase while step frequency will likely decrease (16). In sports, athletes may be required to accelerate from a static position or from a rolling start where they may already be walking or jogging prior to the sprint. Therefore, both of these forms of acceleration should be addressed in speed and agility training.

Agility

The term “agility” encompasses a wide range of movements and abilities that athletes are required to perform throughout competition. Agility movements can be defined as rapid, whole-body movements that require single or multiple changes in velocity (i.e., acceleration or deceleration) or direction (i.e., vertical, lateral, or horizontal) in response to an external stimulus (e.g., opponent movements, location of the ball, etc.) (14,26,27). From this definition, it is clear that reactive decision-making, technique, and physical capabilities all contribute to the success of an agility performance; therefore, agility training programs should include all three of these aspects.

While some training approaches are beneficial for optimal gains in agility performance (i.e., interval training), others such as aerobic conditioning focus on opposing energy systems (oxidative versus phosphagen) and muscle fibers (slow twitch versus fast twitch) than that required for rapid and explosive movements in agility. Coaches should limit high-mileage running (e.g., distance running) for athletes during the base lifting phase and for athletes that are not required to perform aerobically during competition. By including distance running in anaerobic agility programs, strength gains may be compromised, training intensity may decrease due to athletes pacing themselves (as a result of limited rest during training sessions), and the potential for overuse injuries will likely increase. Above all, agility training should be performed in an environment and manner that simulates the conditions and demands of the competition as closely as possible. If the athletes are regularly required to perform lateral cuts followed by short bursts of speed in games or matches, then these movement sequences should be replicated in training. In contrast, if athletes are not required to transition from backpedaling to forward sprinting in their sport, then there is no need to include these tasks as regular elements in the training program.

Interval Training

Interval training is a series of high-intensity, short duration exercises or drills followed by a period of rest. The ratio of this work-to-rest interval is typically 1-to-10 (which is applicable to most power-based sports). This means if the drill lasts 5 s, the athlete should then be given 50 s to recover before performing another drill. Rather than having all athletes in a single group run at the same time, rest cycles can be easily built into drills by dividing the athletes into two smaller groups. This allows one group to perform the drill while the other rests. It is important to

make sure that the resting group is still given their full 50 s of rest before performing the drill again (i.e., the resting group will not be starting the drill immediately following the previous group's completion). The intensity within each drill can be increased to challenge the performance of the athletes. Having athletes race each other is an easy way to increase the intensity of the drill.

A common training error of coaches is making the rest interval too short. For example, when athletes are required to jog or run to the next station with no recovery between drills, athletes tend to pace themselves and decrease the intensity so they are able to complete the workout. Athletes should be giving their best effort for each performance, which means adequate rest and hydration between sets and drills is required.

Components of Agility

There are many factors that contribute to the overall success of a given agility performance. Fortunately, these qualities can be grouped into four main categories: 1) perceptual/decision-making factors; 2) technical factors; 3) physical factors; and 4) anthropometric factors (body shape and size) (Figure 5-3) (17). The following section will briefly describe the importance of each component to agility performance and the trainability of each.

Perceptual Decision-Making Factors

The perceptual/decision-making component relies on the ability of the athlete to recognize specific cues from the situation, and react in a timely and appropriate manner. Decision-making is a difficult quality to train in an athlete due to constantly varying scenarios and circumstances in sports. However, routinely familiarizing players with situations they are likely to encounter in competition will aid in the development of the ability to recognize familiar cues, and anticipate subsequent movements and help determine how to react to such scenarios in competition. While most agility drills do not incorporate a reactive/decision-making component, this perceptual element can easily be integrated into many drills through audio tones (e.g., whistle blow), visual stimuli (e.g., light or color indicators), or the inclusion of additional players (e.g., opponents or teammates).

Technical Factors

The development of proper technique is crucial to the success of many team and individual sports. The same is true for the efficient and effective execution of agility drills. Beginning at the developmental level, proper technique in all aspects of the sport should be emphasized. As athletes progress to more advanced competition and skill levels of the sport, the techniques that were learned early will begin to become autonomous, allowing athletes to shift their focus to other aspects of the sport. As illustrated in Figure 5-3, depending on the specific requirements of the sport, technical characteristics may include body positioning, stride adjustments, foot placement, and joint and body segment sequencing throughout different phases of each movement (17). As the movements performed in the sport should be replicated as much as possible in training sessions, these technical characteristics should also be emphasized in all drills.

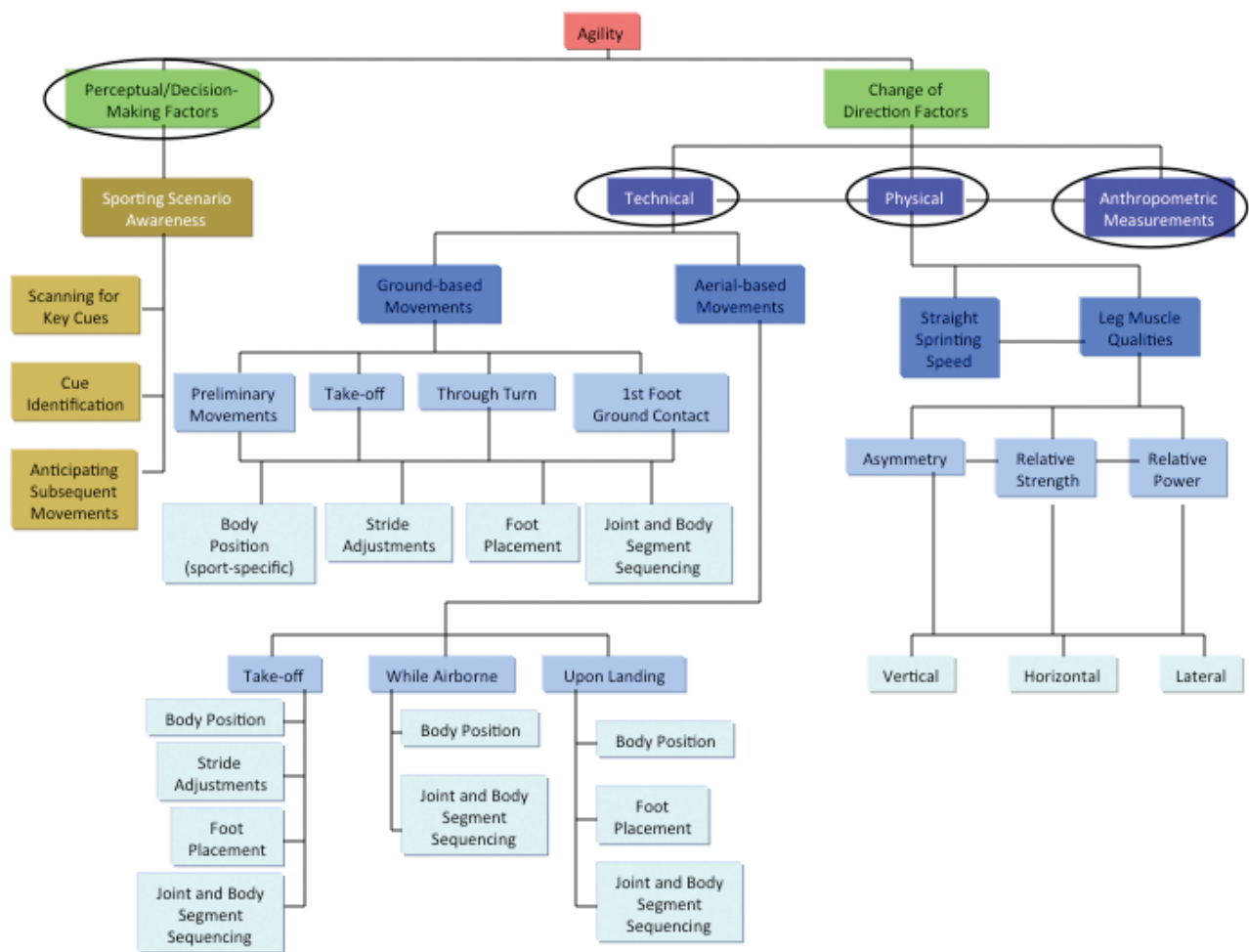


Figure 5-3. Key Contributing Components of Agility (with the four main categories circled) (17)

Athletes from a variety of different sports are required to perform jumps in various directions. While the techniques employed during the approach and take-off phases of a jump are important for attaining maximal height, distances, positioning on the field/court, the techniques used when landing are even more important when trying to minimize the risk of injury.

Noncontact anterior cruciate ligament (ACL) injuries are common among athletes, particularly when landing from a jump. Females sustain noncontact ACL injuries 2 – 8 times more often than males (12,23). This is often attributed to a wider pelvis and increased Q-angle (Figure 5-4) associated with females when compared to males. Increased leg stiffness upon landing coupled with decreased strength in the thigh musculature when compared to males is also likely to contribute to the higher incidence of noncontact ACL injuries in female athletes (23). These anatomical differences and strength deficits typically cause the knees to come together upon landing (valgus knee position) (Figure 5-5), which increases the strain on the ligaments and muscles surrounding the knee.

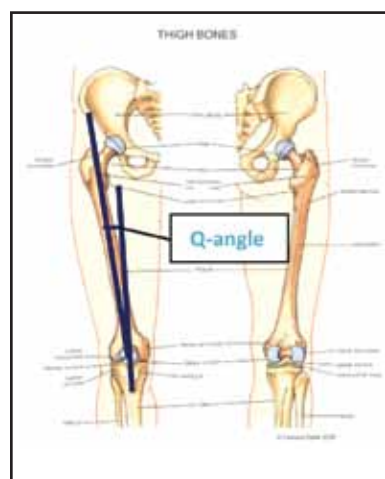


Figure 5-4. The Q-angle

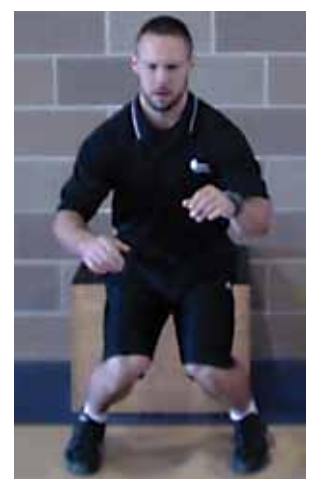


Figure 5-5. Valgus Knee Alignment Upon Landing

It is important to emphasize proper landing technique to athletes regardless of skill level. Joint stiffness upon landing (i.e., limited flexion at the hips, knees, and ankles upon impact) will require the large ground reaction forces to be absorbed by only one or two joints (primarily the knee). Landing on the balls of the feet and immediately flexing the hips, knees, and ankles upon impact will help dissipate the large ground reaction forces over several joints and accompanying muscles, and will likely result in decreased strain on the knees. It is also important to emphasize proper lower body alignment upon landing. The hips, knees, and ankles should create a straight line when viewed from the front (Figure 5-6) throughout the absorption phase of jump landing. This form should also be emphasized when performing a countermovement (rapid, shallow squat) prior to a jump. While some athletes may still incur ACL injuries, emphasizing proper landing mechanics and strengthening the thigh muscles will assist in minimizing the potential for such injuries.



Figure 5-6. Proper Knee Alignment Upon Landing

technique (e.g., arm action, posture, step length, stride frequency, etc.) and overall acceleration performance should be included in the training program prior to the addition of agility drills.

Anthropometric Factors

The overall size and shape of the athlete will also affect agility performance. For example, an athlete with long legs (e.g., basketball center) will naturally have a longer step length than an athlete with shorter legs (e.g., soccer midfielder). When accelerating out of a sharp change of direction, a high step frequency and short step length will allow the athlete to accelerate faster (increased ground contact time). The basketball center will need additional time to reposition his/her longer legs between each step. Therefore, the soccer midfielder will likely have a faster performance time for the first 2 – 3 strides out of a sharp turn. Other factors such as body mass, torso length, and joint flexibility will also have advantages and disadvantages in sports. Some athletes may need to alter their technique in order to capitalize on their natural anthropometric characteristics (17).

Agility Training Drills and Programming

Knowing when to incorporate agility training into a strength training and conditioning program is important for optimal gains in performance. Programming should begin with four weeks of lifting four times per week with no additional running. Beginning on Week 5, speed drills should be added twice per week on nonconsecutive explosive days (see the end of this chapter for sample speed programs). Agility drills can be added to the program on the strength days beginning on Week 7 (see the end of this chapter for sample agility programs). These two strength days should be opposite to the explosive days, making four running days per week.

As with any training program, all exercises and skills should be progressive in nature. The first time athletes perform a new drill, they will not be familiar with the movement. A significant amount of attention by the athletes will be required to perform the drill. Once the athletes have become familiar with the drill (familiarity will vary depending on the difficulty of the drill) they will know some key aspects of the movement, but attention will still be needed when performing the drill. When practiced extensively, the athletes will develop better knowledge of the movement and will be able to perform the drill with less conscious attention to the specific movement patterns. At this point, additional challenges (e.g., more complex movement patterns, catching a ball, etc.) may be incorporated into the drill to increase the difficulty. It is important to allow athletes to progress through all three phases of development before increasing the complexity of the drill for optimal performance gains.

The following information pertains to standard techniques for basic agility training drills and some of their variations. Many strength training and conditioning programs regularly use these foundational drills. Everyone using these drills should have a sound understanding of how each of these drills and their variations are performed to optimize individual techniques. Performance technique for each drill listed below appears in the remainder of this chapter. Examples of agility training programs are provided at the end of this chapter.

Warm-Up Drills

1. High Knees – 10 yards down and back
2. Heel-Ups – 10 yards down and back
3. Forward Lunge with Elbow to Instep – 10 yards
4. Side Lunge with Squat – 4 each side
5. High Knee Foreleg Extension – 10 yards down slow, 10 yards back quick

Speed Drills

1. Build-Ups – 40 yards
2. Form Starts
3. Position Starts
4. Flying 10s
5. Power Skips (for height)
6. Power Skips (for distance)
7. Flying 20s
8. Harness Routine
9. Flying 30s
10. Bag Jumps
11. Hollow Sprints

Agility Drills

1. Rope or Ladder Routine
 - 1a. Every Hole
 - 1b. Every Other Hole
 - 1c. Lateral Step
2. Bag Routine
 - 2a. Change of Direction

- 2b. Shuffle
- 2c. Forward and Backpedal

3. Line Jump Routine
 - 3a. Single Bunny Hop
 - 3b. Double Bunny Hop
 - 3c. Scissors
 - 3d. Ali shuffle

4. Pro-Agility
5. Nebraska Agility
6. Three-Cone Drill
7. Four-Corner Drill
8. Sprint Ladder
9. Shuffle Ladder
10. Backpedal Ladder

Landing Drills

1. Drop Jump
2. Vertical Jump
3. Tuck Jump
4. 180 Degree Jump
5. Broad Jump with Vertical Jump
6. Depth Jump
7. Box Shuffle Step
8. Double Box Shuffle Step
9. Lateral Box Jump

Warm-Up Drills

1. High Knees – 10 yards down and back

Exercise Objective: Develop lower body muscle strength, power, and increased range of motion of the hips

Procedure

1. Forcefully drive knee upward (Figure 5-7)
2. Fully extend opposite leg as driving knee is lifted
3. Maintain a flat back with a slight forward lean of the torso
4. Drive arms aggressively in opposite direction of knee drive



Figure 5-7. High Knees



Figure 5-8. Heel-Ups

Coaching Points

- Take short, quick steps (the athlete should achieve at least 30 steps over 10 yards)
- Avoid leaning back
- Thigh should be parallel to the ground at the highest point of the movement
- Keep head and neck relaxed throughout the movement

2. Heel-Ups – 10 yards down and back

Exercise Objective: Develop hamstring strength and range of motion of the hips

Procedure

1. Swing heel up toward buttocks with each step (Figure 5-8)

2. Knee of swinging leg should remain in line with the body (not flexing or extending at the hip)
3. Maintain a flat back with a slight forward lean of the torso
4. Move arms in opposition to legs but keep shoulders relaxed (no driving action of the arms)

Coaching Points

- Take short, quick steps
- Avoid leaning back
- Weight should remain on the forefoot throughout the movement
- Keep head, neck, and arms relaxed throughout the movement

3. Forward Lunge with Elbow to Instep – 10 yards

Exercise Objective: Develop strength and range of motion of the hips

Procedure

1. Take a large step forward with the left foot (Figure 5-9a)
2. Bring torso down toward front knee (using the right hand to help balance if needed) (Figure 5-9b)
3. Bring left elbow towards inside of left foot (Figure 5-9c) and hold for 3 s
4. Push off with left foot and bring right leg forward past left leg, and repeat movement on right side of the body

Coaching Points

- Keep front foot pointed forward and feet hip-width apart
- When bringing the torso down, the knee should remain directly above the ankle
- No additional steps should be taken when transitioning from the left to right side

4. Side Lunge with Squat – 4 each side

Exercise Objective: Develop strength and range of motion of the hips

Procedure

1. Begin with feet shoulder-width apart with knees and hips slightly flexed
2. Take a large step to the left, keeping both feet facing forward
3. Push hips back, shift weight onto left leg, and flex left knee and hip (right leg remains straight) (Figure 5-10a)
4. Hold position for 3 s



Figure 5-9a. Forward Lunge



Figure 5-9b. Bring Torso Toward Front Knee



Figure 5-9c. Bring Elbow Toward Instep

5. Stand up through left leg by keeping weight on left leg and bringing right leg back to shoulder-width
6. Lower into a squat, keeping weight on heels (Figure 5-10b)
7. Return to start position

Coaching Points

- When shifting weight onto left leg, the knee should remain directly above the ankle
- Keep right foot flat on the ground while shifting weight to left leg

5. High Knee Foreleg Extension – 10 yards down slow, 10 yards back fast

Exercise Objective: Develop strength and range of motion of the hips and knees

Procedure

1. Perform a small skip by driving left knee up and extending onto toes of right foot (Figure 5-11a)
2. At the highest point of the skip, extend left knee until entire leg is parallel to the ground (Figure 5-11b)
3. Keep right knee extended and drive left leg down toward the ground

Coaching Points

- Maintain a slight forward lean throughout
- The thigh should reach parallel to the ground before extending the foreleg
- Try to make the movement rhythmic

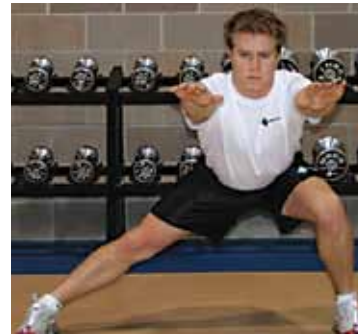


Figure 5-10a. Side Lunge Position

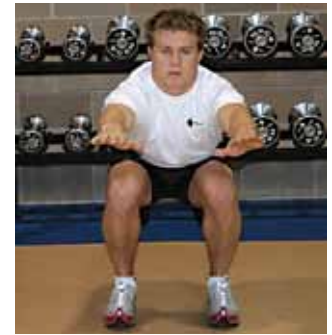


Figure 5-10b. Squat Position



Figure 5-11a. Driving the Knee Upward



Figure 5-11b. Knee Extension

Speed Drills

1. Build-Ups – 40 yards

Exercise Objective: Develop proper stride length and stride frequency

Procedure

1. Begin jogging from a standing start
2. Gradually build up speed so that maximum velocity is attained by 40 yards (Figure 5-12)
3. Gradually slow the pace over the remaining 20 yards

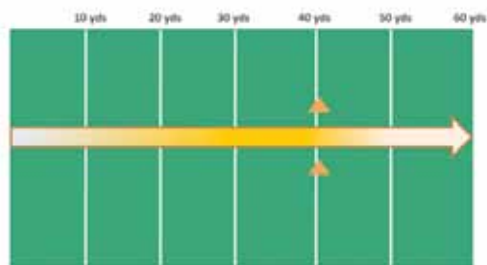


Figure 5-12. 40-Yard Build-Up

Coaching Points

- Avoid running at full speed after 40 yards
- Make sure proper sprinting technique is employed throughout the drill

2. Form Starts

Exercise Objective: Develop proper stride length, stride frequency, and body position from a static start

Procedure

1. Place one foot directly behind the designated start line
2. Place other foot 6 – 12 in. behind heel of front foot and 2 – 4 in. to the inside of front foot (Figure 5-13)
3. The hand opposite to the front foot is placed just behind the start line (usually on fingertips)
4. The other arm is extended behind body with a bend at the elbow
5. Hips should be higher than shoulders
6. A slight forward lean is used, which can cause the shoulders to cross over the start line
7. Eyes should be focused 2 – 3 ft in front of the start line



Figure 5-13. Form Start

Coaching Points

- Push off with both feet at the start of the sprint
- Maintain a forward lean with head down for the first 10 yards, then gradually become more erect through the next 10 yards
- This start can be used for any sprinting distance of interest during the training session

3. Position Starts

Exercise Objective: Develop proper stride length, stride frequency, and body position from a static start

Procedure

1. Begin in the Athletic Stance (Figure 5-14) with hips, knees, and ankles slightly flexed, and hands in front of the body
2. On cue, explode out of this stance into the required direction and distance as specified by the coach or drill

Coaching Points

- Remain in the athletic stance until the cue is given
- The initial step out of the stance should be powerful and explosive



Figure 5-14. Athletic Stance

4. Flying 10s

Exercise Objective: Develop proper stride length and stride frequency

Procedure

1. Begin jogging at half speed, increasing speed with each stride for the first 30 yards so maximal velocity is attained by the 30-yard mark (Figure 5-15)
2. Continue to sprint at maximum velocity for 10 yards before decelerating gradually

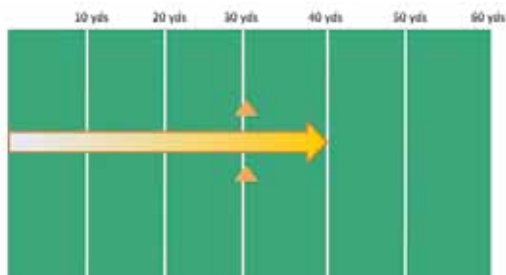


Figure 5-15. Flying 10s

Coaching Points

- A maximal sprint should be performed for the entire 10 yards
- Make sure proper sprinting technique is employed throughout the drill

5. Power Skips (for height)

Exercise Objective: Develop explosive power in the lower body



Figure 5-16. Power Skips for Height

Procedure

1. Perform a high skip by driving the left knee up, maintaining a 90° angle at the knee, landing on the take-off leg (the extended leg)
2. Try to attain maximal height during every skip (Figure 5-16)
3. Drive arms aggressively in opposition to the legs

Coaching Points

- The take-off leg should reach full extension at the ankle, knee, and hip for every skip
- The goal should be to skip as high as possible, not as far as possible

6. Power Skips (for distance)

Exercise Objective: Develop explosive power in the lower body

Procedure

1. Perform a high skip by driving the left knee forward and upward, maintaining a 90° angle at the knee, landing on the take-off leg (the extended leg)
2. Try to attain maximal distance during every skip (Figure 5-17)
3. Drive arms aggressively in opposition to the legs



Figure 5-17. Power Skips for Distance

Coaching Points

- The take-off leg should reach full extension at the ankle, knee, and hip for every skip
- The goal should be to skip as far as possible, not as high as possible

7. Flying 20s

Exercise Objective: Develop proper stride length and stride frequency

Procedure

1. Begin jogging at half speed, increasing speed with each stride for the first 30 yards so maximal velocity is attained by the 30-yard mark (Figure 5-18)
2. Continue to sprint at maximum velocity for 20 yards before decelerating gradually

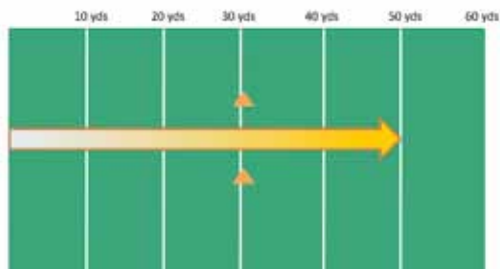


Figure 5-18. Flying 20s

Coaching Points

- A maximal sprint should be performed for the entire 20 yards
- Make sure proper sprinting technique is employed throughout the drill

8. Harness Routine

Exercise Objective: Develop proper body positioning during acceleration phase and minimize the time needed to attain maximum velocity

Procedure

1. Attach harness around waist with handles or slack in the back (Figure 5-19)
2. Attach back of harness to stationary object or have coach hold
3. Sprint in place with the following technique:
 - A. Drive lead leg upwards so thigh is parallel to ground
 - B. Fully extend trail leg at ankle, knee, and hip
 - C. Drive arms aggressively in opposition to the legs
 - D. Elbows should flex to 90° on forward arm swings and fully extend on backward arm swings
 - E. Arms should remain parallel to the body (i.e., not crossing over in front of the chest)
 - F. A slight forward lean into the harness is used throughout the drill
4. After sprinting in place for 10 s, the same technique should be used with the harness for the remaining tasks (i.e., 10-yard sprint, 10-yard jog/10-yard sprint) to complete the repetition



Figure 5-19. Sprinting with a Harness

Coaching Points

- Maintain proper body lean and posture throughout each task
- Enforce appropriate rest intervals throughout the drill

**NOTE: One repetition includes sprinting in place, sprinting 10 yards, jogging 10 yards followed immediately by sprinting 10 yards (an area with 50 – 60 yards of flat running surface is needed)*

9. Flying 30s

Exercise Objective: Develop appropriate stride length and stride frequency

Procedure

1. Begin jogging at half speed, increasing speed with each stride for the first 30 yards so maximal velocity is attained by the 30-yard mark (Figure 5-20)
2. Continue to sprint at maximum velocity for 30 yards before decelerating gradually

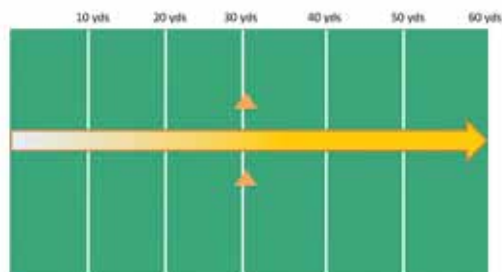


Figure 5-20. Flying 30s

Coaching Points

- A maximal sprint should be performed for the entire 30 yard sprint
- Make sure proper sprinting technique is employed throughout the drill

10. Bag Jumps

Exercise Objective: Develop explosive power and acceleration



Figure 5-21. Bag Jump Tuck Form

Procedure

1. A series of bags (approximately 24 in. high) are laid out in a line on the ground with approximately 24 in. between bags
2. Jump over the first bag, tucking both knees in toward the chest (Figure 5-21)
3. Drive both arms upward when exploding off of the ground to assist in reaching optimal jump height
4. Upon landing, immediately explode upward over the next bag in the series
5. Sprint 5 yards following the final bag jump

Coaching Points

- Minimize time on the ground by quickly exploding into the next bag jump
- Avoid taking additional steps between bags
- Make sure both feet are brought over the bags and not around the sides

11. Hollow Sprints

Exercise Objective: Develop proper stride length and stride frequency, as well as explosive power

Procedure

1. Perform the following intervals consecutively without stopping (Figure 5-22):
 - A. Jog for 10 yards
 - B. Sprint for 10 yards
 - C. Jog for 10 yards
 - D. Sprint for 10 yards
 - E. Jog for 10 yards
2. This drill can also be performed using 20-yard increments (100 yards total for the drill) and 30-yard increments (150 yards total for the drill)

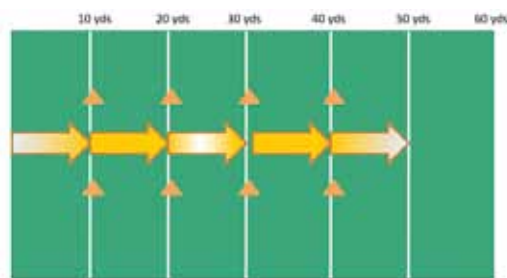


Figure 5-22. Hollow Sprints (10-yard increments shown)

Coaching Points

- Smoothly transition between running speeds
- When decelerating to a slower speed, sit into a shallow (three quarter) squat as opposed to sticking the chest out
- Sprint at full speed for the entire designated distance before decelerating

Agility Drills

1. Rope or Ladder Routine

Exercise Objective: Develop range of motion of the hips, knees, and ankles, as well as coordination and balance

1a. Every Hole

Procedure

1. Run forward, stepping into every hole with each foot (Figure 5-23)
2. Drive knees upward for each step
3. Drive arms aggressively in opposition to the legs
4. A slight forward lean is used throughout the drill

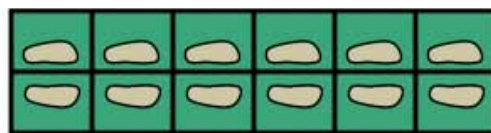


Figure 5-23. Rope or Ladder Routine - Every Hole

Coaching Points

- The right foot hits every hole on the right side; the left foot hits every hole on the left side
- Keep head up with eyes focused straight ahead (avoid looking down)

1b. Every Other Hole

Procedure

1. Run forward, stepping into every other hole with each foot (Figure 5-24)
2. Drive knees upward for each step
3. Drive arms aggressively in opposition to the legs
4. A slight forward lean is used throughout the drill

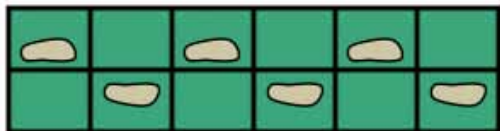


Figure 5-24. Rope or Ladder Routine – Every Other Hole

Coaching Points

- The right foot hits every other hole on the right side; the left foot hits every other hole on the left side
- Keep head up with eyes focused straight ahead (avoid looking down)

1c. Lateral Step

Procedure

1. Run laterally, stepping into every hole with each foot (Figure 5-25)
2. Use only one side of the ladder per direction
3. Drive knees upward for each step
4. Drive arms aggressively in opposition to the legs
5. A slight forward lean is used throughout the drill

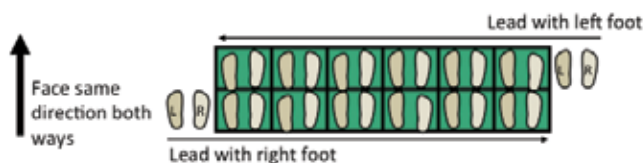


Figure 5-25. Rope or Ladder Routine – Lateral Step

Coaching Points

- Face the same direction for each leading foot
- Keep head up with eyes focused straight ahead

2. Bag Routine

Exercise Objective: Develop range of motion of the hips, knees, and ankles, as well as coordination and balance when accelerating, decelerating, and changing directions rapidly

2a. Change of Direction

Procedure

1. Position bags 5 yards apart
2. Beginning at either the right or left bag, run forward to the far side of the first bag (Figure 5-26)
3. Use the outside foot to plant and run forward toward the far side of the next bag (continue this weaving pattern through the three bags)
4. Finish the drill by sprinting 5 yards

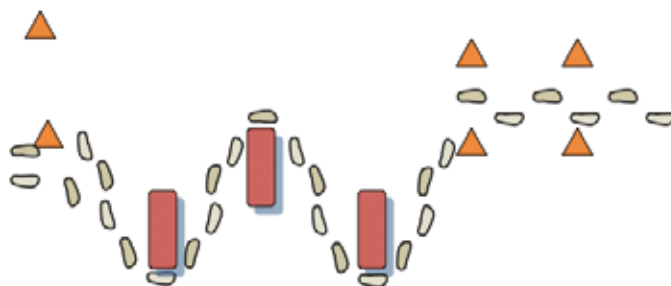


Figure 5-26. Bag Routine – Change of Direction

Coaching Points

- Drive knees upward for each step
- Drive arms aggressively in opposition to the legs
- A slight forward lean is used throughout the drill
- Take only one step around each bag

2b. Shuffle

Procedure

1. Position bags 5 yards apart
2. Beginning at either the right or left bag, shuffle diagonally to the far side of the first bag (Figure 5-27)
3. Use the outside foot to plant and explode diagonally toward the far side of the next bag (continue this weaving pattern through the three bags)
4. Finish the drill by sprinting 5 yards

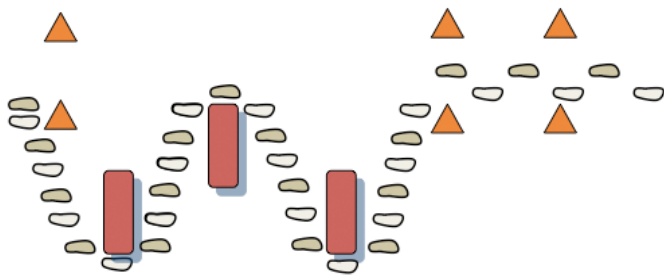


Figure 5-27. Bag Routine - Shuffle

Coaching Points

- Stay low throughout the drill
- Avoid using a cross-over step
- Take only one step around each bag

2c. Forward and Backpedal

Procedure

1. Beginning at either the right or left bag in the athletic position (upright torso, slight flexion of the ankles, knees and hips, head up and hands away from the body)
2. On cue, backpedal to the far end of the first bag (Figure 5-28)
3. Transition into a forward sprint at the far end of the next bag (continue this pattern through the remaining bags)
4. Finish the drill by sprinting 5 yards

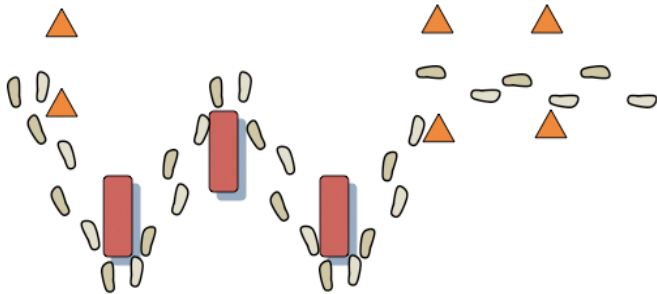


Figure 5-28. Bag Routine - Forward and Backpedal

Coaching Points

- Stay low throughout the drill
- Keep weight on the balls of the feet when backpedaling
- Avoid looking back for the bags
- Take only one step around each bag

3. Line Jump Routine

Exercise Objective: Develop range of motion of the ankles, as well as lower body strength, coordination, and balance when changing directions rapidly

3a. Single Bunny Hop

Procedure

1. Stand at a straight line on a field or court
2. Jump back and forth on one foot over the line, moving forward (Figure 5-29)
3. Switch feet at the midway point without stopping



Figure 5-29. Line Jump Routine - Single Bunny Hop

Coaching Points

- Stay as close to the line as possible
- Keep head up with eyes focused straight ahead
- This drill can also be performed moving backwards

3b. Double Bunny Hop

Procedure

1. Stand with both feet to one side of a straight line on a field or court
2. Moving forward, jump back and forth over the line with both feet (Figure 5-30)



Figure 5-30. Line Jump Routine - Double Bunny Hop

Coaching Points

- Stay as close to the line as possible
- Keep head up with eyes focused straight ahead
- This drill can also be performed moving backwards

3c. Scissors

Procedure

1. Stand with one foot on each side of a straight line on a field or court
2. Moving forward, shuffle step down the line (Figure 5-31)

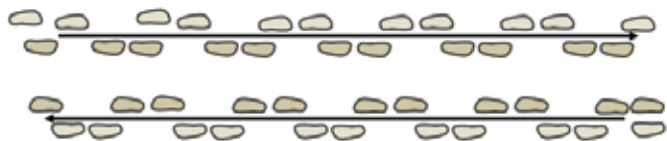


Figure 5-31. Line Jump Routine - Scissors

Coaching Points

- Stay as close to the line as possible
- Keep head up with eyes focused straight ahead

3d. Ali Shuffle

Procedure

1. Stand with one foot on either side of a line on a field or court (toes of the right foot to the line and the heel of the left foot to the line) (Figure 5-32)
2. Moving laterally down the line, switch feet while jumping in the air (the right foot will now be on the far side of the line with the heel nearest the line, and the left foot will now be on the near side with toes to the line)

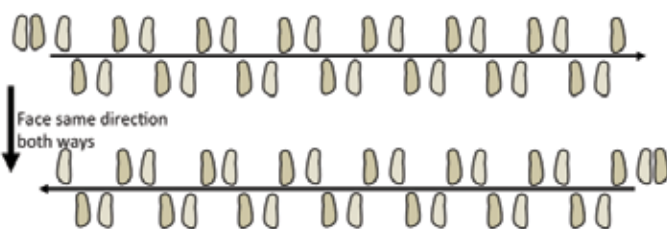


Figure 5-32. Line Jump Routine - Ali Shuffle

Coaching Points

- Stay low to the ground while jumping
- Keep head up with eyes focused straight ahead
- Shuffle to the right and then to the left



Figure 5-33a. Starting Position for the Pro-Agility Drill

4. Pro-Agility

Exercise Objective: Improve reaction time, coordination, balance, and acceleration when changing directions rapidly

Procedure

1. Begin in the athletic stance, straddling the middle line (Figure 5-33a)
2. When ready, sprint to the right line and touch it with the right hand
3. Push off laterally, sprint back across the middle line to the left line, and touch it with the left hand
4. Push off laterally and sprint back through the middle line (Figure 5-33b)

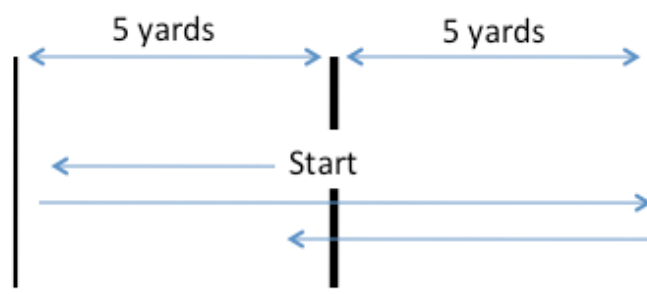


Figure 5-33b. Pro-Agility Drill

Coaching Points

- Stay low when changing directions
- Make sure to touch the lines with the corresponding hand before changing directions (do not just touch the line with foot)

5. Nebraska Agility

Exercise Objective: Improve coordination, balance, and acceleration when changing directions rapidly

Procedure

1. Begin in the Form Start position (Figure 5-13) at the starting line (Figure 5-34)
2. When ready, sprint forward to the cone and plant left foot to complete a 180° change of direction to the right
3. Sprint forward back to the starting line and plant right foot to complete a 180° change of direction to the left

4. Sprint forward 5 yards and touch the line with both hands; then backpedal across the starting line

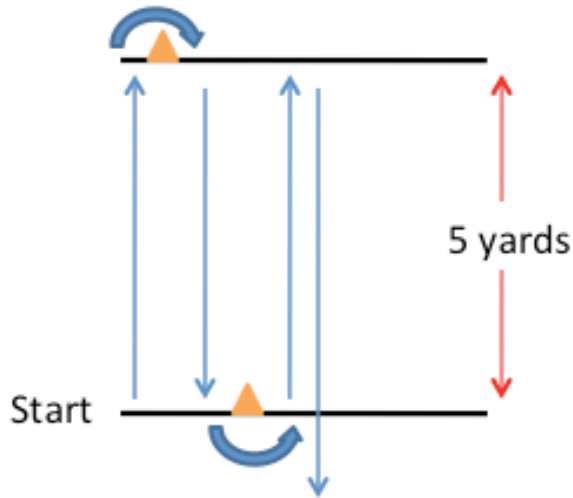


Figure 5-34. Nebraska Agility Drill

Coaching Points

- Stay low when changing directions and backpedaling
- Avoid using more than one step to change directions around the cones
- Make sure to touch the line with both hands before backpedaling

6. Three-Cone Drill

Exercise Objective: Improve coordination, balance, acceleration, and deceleration when changing directions rapidly

Procedure

1. Begin in the Form Start position (Figure 5-13) at the starting line (Figure 5-35)
2. When ready, sprint forward to the cone and plant right foot to complete a 90° change of direction to the left
3. Sprint forward to the far side of the next cone; plant left foot to complete a 180° change of direction to the right
4. Sprint forward back to the second cone and plant left foot to complete a 90° change of direction to the right
5. Sprint past the first cone to finish
6. Complete this drill in the opposite direction as well (i.e., right turn around the first cone, etc.)

Coaching Points

- Stay low when changing directions

- Avoid using more than one step to change directions around the cones
- Keep head and eyes up; avoid looking for the cones

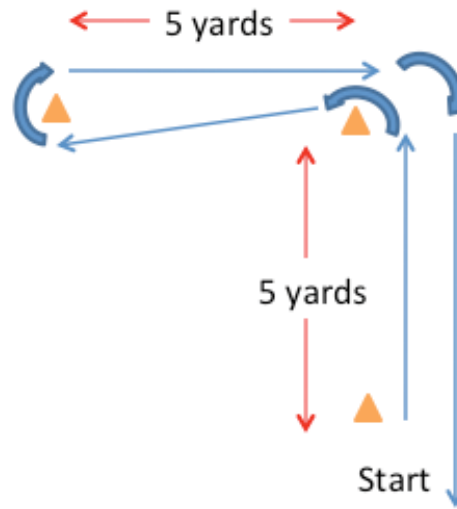


Figure 5-35. Three-Cone Drill

7. Four-Corner Drill

Exercise Objective: Improve coordination, balance, acceleration, and deceleration when changing directions rapidly

Procedure

1. Begin in the Form Start position (Figure 5-13) at the starting line (Figure 5-36)
2. When ready, sprint forward to the cone and plant right foot to complete a 90° change of direction to the left
3. Sprint forward to the next cone and plant right foot to complete a 90° change of direction to the left
4. Sprint forward to the next cone and plant right foot to complete a 90° change of direction to the left
5. Sprint past the first cone to finish
6. Complete this drill in the opposite direction as well (i.e., plant with left foot to complete right turns around all cones)
7. This drill can also be performed using a sprint-shuffle-backpedal-shuffle pattern and a backpedal-shuffle-sprint-shuffle pattern

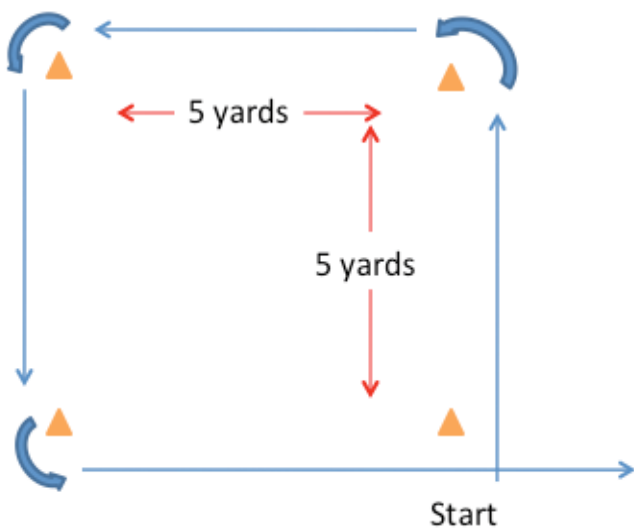


Figure 5-36. Four-Corner Drill

Coaching Points

- Stay low when changing directions
- Avoid using more than one step to change directions around the cones
- Keep head and eyes up; avoid looking for the cones

8. Sprint Ladder

Exercise Objective: Improve coordination, balance, acceleration, and deceleration when changing directions rapidly

Procedure

1. Begin in the Form Start position (Figure 5-13) at the starting line
2. When ready, sprint forward to the first line (5 yards) and touch it with the right hand (Figure 5-37)
3. Turn and sprint back to the starting line and touch it with the left hand
4. Turn and sprint to the second line (10 yards) and touch it with the right hand
5. Turn and sprint back to the starting line and touch it with the left hand
6. Turn and sprint to the first line again (5 yards), touch it with the right hand, then turn and sprint past the starting line
7. Perform this drill using the following distances: 10 yards – 5 yards – 10 yards, 5 yards – 5 yards – 5 yards, 5 yards – 10 yards – 15 yards, and 15 yards – 10 yards – 5 yards

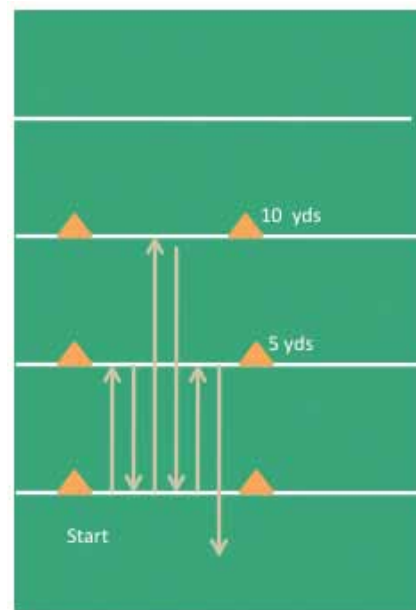


Figure 5-37. Sprint Ladder

Coaching Points

- Stay low when changing directions
- Avoid using more than one step to change directions
- Keep turns sharp – avoid running in a circular pattern when turning

9. Shuffle Ladder

Exercise Objective: Improve coordination, balance, and strength in the adductors and abductors

Procedure

1. Begin in the Athletic Stance (Figure 5-14) straddling the starting line
2. When ready, shuffle to the first line (5 yards) and touch it with the outside hand (i.e., if facing left, touch the line with the right hand) (Figure 5-38)
3. Shuffle back to the starting line and touch it with the outside hand
4. Shuffle to the second line (10 yards) and touch it with the outside hand
5. Shuffle back to the starting line and touch it with the outside hand
6. Shuffle to the first line again (5 yards), touch it with the outside hand then shuffle past the starting line
7. Perform this drill facing both the right and the left using the following distances: 10 yards – 5 yards – 10 yards, 5 yards – 5 yards – 5 yards, 5 yards – 10 yards – 15 yards, and 15 yards – 10 yards – 5 yards



Figure 5-38. Touching the Line with the Outside Hand

Coaching Points

- Stay low when shuffling
- Touch the line with the outside hand

10. Backpedal Ladder

Exercise Objective: Improve coordination, balance, acceleration, and deceleration ability when changing directions rapidly

Procedure

1. Begin in the Athletic Stance (Figure 5-14) facing away from the starting line
2. When ready, backpedal to the first line (5 yards) and touch it with either foot
3. Turn and backpedal back to the starting line and touch it with both hands
4. Turn and backpedal to the second line (10 yards) and touch it with either foot
5. Turn and backpedal back to the starting line and touch it with both hands
6. Turn and backpedal to the first line again (5 yards), touch it with either foot, then turn and backpedal past the starting line
7. Perform this drill using the following distances: 10 yards – 5 yards – 10 yards, 5 yards – 5 yards – 5 yards, 5 yards – 10 yards – 15 yards, and 15 yards – 10 yards – 5 yards

Coaching Points

- Stay low when backpedaling
- Avoid using more than one step to change directions

Landing Drills

1. Drop Jump

Exercise Objective: Develop proper landing technique for absorbing large impact forces

Procedure

1. Stand on a box (12 – 24 in. tall) with feet hip-width apart and toes pointed forward (Figure 5-39a)
2. Step off the box and land on the balls of both feet simultaneously
3. Flex hips, knees, and ankles immediately upon impact (Figure 5-39b)
4. Hold this flexed position for 5 s



Figure 5-39a. Start Position



Figure 5-39b. Landing Position

Coaching Points

- Avoid jumping off the box
- The landing should be soft and quiet
- Knees should be directly over ankles throughout the landing (avoid letting knees come together upon impact)



Figure 5-40. Vertical Jump

2. Vertical Jump

Exercise Objective: Develop explosive power

Procedure

1. Stand with feet hip-width apart and toes pointed forward
2. Lower 4 – 6 in. (i.e., countermovement) and immediately extend hips, knees, and ankles while driving both arms upward, jumping as high as possible (Figure 5-40)
3. Land softly by flexing hips, knees, and ankles upon impact

Coaching Points

- Avoid taking any steps before the jump
- There should be no pause between lowering and extending
- The landing should be soft and quiet
- The knees should be directly over the ankles throughout the landing (avoid letting the knees come together upon impact)

3. Tuck Jump



Figure 5-41. Tuck Jump

Exercise Objective: Develop explosive power, and develop proper landing technique for absorbing large impact forces

Procedure

1. Stand with feet hip-width apart and toes pointed forward
2. Lower 4 – 6 in. (i.e., countermovement) and immediately extend hips, knees, and ankles
3. While airborne, bring both knees into the chest (Figure 5-41)
4. Land softly by flexing hips, knees, and ankles upon impact

Coaching Points

- Avoid taking any steps before the jump
- There should be no pause between lowering and extending
- The landing should be soft and quiet
- Knees should be directly over ankles throughout the landing (avoid letting the knees come together upon impact)

4. 180 Degree Jump

Exercise Objective: Develop explosive power, and develop proper landing technique for absorbing large impact forces

Procedure

1. Stand with feet hip-width apart and toes pointed forward
2. Lower 4 – 6 in. (i.e., countermovement) (Figure 5-42a) and immediately extend hips, knees, and ankles while driving both arms upward
3. While airborne, rotate 180° to face the opposite direction (Figure 5-42b)
4. Land softly by flexing hips, knees, and ankles upon impact (Figure 5-42c)

Coaching Points

- Avoid taking any steps before the jump
- There should be no pause between lowering and extending
- The landing should be soft and quiet
- Knees should be directly over ankles throughout the landing (avoid letting the knees come together upon impact)



Figure 5-42a. Countermovement Phase of the 180° Jump



Figure 5-42b. Aerial Turn for 180° Jump



Figure 5-42c. 180° Jump Landing

5. Broad Jump with Vertical Jump

Exercise Objective: Develop explosive power, and develop proper landing technique for absorbing large impact forces

Procedure

1. Stand with feet hip-width apart and toes pointed forward
2. Lower 4 – 6 in. (i.e., countermovement) while bringing both arms back behind the body (Figure 5-43a)
3. Immediately extend hips, knees, and ankles while driving both arms up and forward, (jumping forward at 45°) (Figure 5-43b)
4. Bring both legs forward to land with both feet ahead of hips (Figure 5-43c)
5. Repeat the broad jump (forward jump) three times, and then perform one vertical jump for maximum height (Figure 5-43d)

Coaching Points

- Avoid taking any steps between jumps
- There should be no pause between lowering and extending
- The landing should be soft and quiet
- Knees should be directly over ankles throughout the landing (avoid letting the knees come together upon impact)



Figure 5-43a. Countermovement Phase of the Broad Jump



Figure 5-43b. Broad Jump Takeoff



Figure 5-43c. Broad Jump Landing



Figure 5-43d. Vertical Jump following Broad Jump

6. Depth Jump

Exercise Objective: Develop explosive power, and develop proper landing technique for absorbing large impact forces

Procedure

1. Begin standing on a box (12 – 24 in. tall) with feet hip-width apart and toes pointed forward (Figure 5-44a)
2. Step off the box and land on the balls of both feet simultaneously
3. Flex hips, knees, and ankles immediately upon impact (Figure 5-44b)
4. Rapidly extend hips, knees, and ankles while driving both arms upward, jumping as high as possible (Figure 5-44c)



Figure 5-44a. Depth Jump Start Position



Figure 5-44b. Depth Jump Countermovement



Figure 5-44c. Vertical Jump Portion

Coaching Points

- Avoid jumping off the box
- Avoid pausing between the landing and vertical jump
- The landing should be soft and quiet
- Knees should be directly over ankles throughout the landing (avoid letting the knees come together upon impact)

7. Box Shuffle Step

Exercise Objective: Develop explosive lateral power

Procedure

1. Stand with the left foot on a box (12 – 24 in. tall) with feet hip-width apart and toes pointed forward (Figure 5-45a)
2. Jump up and over the box (Figure 5-45b) with the right foot landing on the far side of the top of the box and the left foot landing on the ground (Figure 5-45c)
3. Shuffle up and over the box from side to side continuously for the designated time (e.g., 20 s)

Coaching Points

- Avoid pausing between the landings
- The landing should be soft and quiet
- Stay low throughout drill



Figure 5-45a. Box Shuffle Step Start Position



Figure 5-45b. Box Shuffle Step Jump



Figure 5-45c. Box Shuffle Step Landing

8. Double Box Shuffle Step

Exercise Objective: Develop explosive lateral power

Procedure

1. Stand to the right side of a box (12 – 24 in. tall) with feet hip-width and toes pointed forward (Figure 5-46a)
2. Jump up onto the box, landing with the left foot (Figure 5-46b)
3. Switch feet on top of the box, so the right foot is now on the box and the left leg is free (Figure 5-46c)
4. Lower the left foot down to the ground (Figure 5-46d)
5. Jump up and over the box from side to side continuously for the designated time (e.g., 20 s)



Figure 5-46a. Double Box Shuffle Step Takeoff



Figure 5-46b. Double Box Shuffle Step First Contact



Figure 5-46c. Double Box Shuffle Step Second Contact



Figure 5-46d. Double Box Shuffle Step Landing

Coaching Points

- Avoid pausing between the landings
- The landings should be soft and quiet
- Stay low throughout drill

9. Lateral Box Jump

Exercise Objective: Develop explosive lateral power

Procedure

1. Stand to one side of a box (12 – 24 in. tall) with feet hip-width and toes pointed forward (Figure 5-47a)
2. Jump up onto the box with both feet (Figure 5-47b)
3. Jump off of the box, landing on the other side with both feet (Figure 5-47c)
4. Jump up and over the box from side to side continuously for the designated time (e.g., 20 s)

Coaching Points

- Avoid pausing between the landings
- The landing should be soft and quiet
- Stay low throughout drill



Figure 5-47a. Lateral Box Jump Start Position



Figure 5-47b. Landing on the Box



Figure 5-47c. Lateral Box Jump Landing (far side)

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Sample Program for Agility Drills Weeks 5-12

Week 5		
Drill	Day 1 Reps (LIGHT)	Day 2 Reps (HEAVY)
Build-Ups	2	4
Form Starts	4	4
Position Starts	4	4
2 minute water break		
Flying 10s	2	3
Power Skips (height)	2	2
Power Skips (distance)	2	2
Total reps	16	19

Week 6		
Drill	Day 1 Reps (LIGHT)	Day 2 Reps (HEAVY)
Build-Ups	3	4
Form Starts	4	4
Position Starts	4	4
2 minute water break		
Flying 10s	3	4
Power Skips (height)	2	3
Power Skips (distance)	2	3
Total reps	18	22

Week 7		
Drill	Day 1 Reps (LIGHT)	Day 2 Reps (HEAVY)
Build-Ups	4	4
Form Starts	4	4
Position Starts	4	4
2 minute water break		
Flying 20s	2	3
Power Skips (height)	2	2
2 minute water break		
Power Skips (distance)	2	2
Harness Routine	1	2
Total reps	21	25

Week 8		
Drill	Day 1 Reps (LIGHT)	Day 2 Reps (HEAVY)
Build-Ups	4	4
Form Starts	4	4
Position Starts	4	4
2 minute water break		
Flying 20s	2	4
Power Skips (height)	2	3
2 minute water break		
Power Skips (distance)	2	3
Harness Routine	2	2
Total reps	24	28

Week 9		
Drill	Day 1 Reps (LIGHT)	Day 2 Reps (HEAVY)
Build-Ups	4	4
Form Starts	4	4
Position Starts	4	4
2 minute water break		
Flying 30s	2	3
Bag Jumps	2	4
2 minute water break		
Harness Routine	2	6
Total reps	22	25

Week 10		
Drill	Day 1 Reps (LIGHT)	Day 2 Reps (HEAVY)
Build-Ups	4	4
Form Starts	4	4
Position Starts	4	4
2 minute water break		
Flying 30s	2	4
Bag Jumps	3	4
2 minute water break		
Harness Routine	2	3
Total reps	23	29

Week 11		
Drill	Day 1 Reps (LIGHT)	Day 2 Reps (HEAVY)
Build-Ups	4	4
Form Starts	4	4
Position Starts	4	4
2 minute water break		
Flying 30s	3	4
Bag Jumps	3	4
2 minute water break		
Harness Routine	2	3
Total reps	24	29

Week 12		
Drill	Day 1 Reps (LIGHT)	Day 2 Reps (HEAVY)
Build-Ups	4	4
Form Starts	4	4
Position Starts	4	4
2 minute water break		
Flying 30s	4	4
Bag Jumps	3	4
2 minute water break		
Harness Routine	2	4
Total reps	25	32

Sample Program for Speed Drills Weeks 7-12

Week 7		
Drill	Day 1 Reps (HEAVY)	Day 2 Reps (LIGHT)
Rope or Ladder Routine (a, b, and c)	1 each	1 each
Bag Routine (a, b, and c)	1 each	1 each
Line Jump Routine (a, b, c, and d)	1 each	1 each
2 minute water break		
Pro-Agility	4 (2 left, 2 right)	2 (1 left, 1 right)
Three-Cone Drill	4 (2 left, 2 right)	4 (2 left, 2 right)
2 minute water break		
Sprint Ladder	4 (1 x a, 1 x b, 1 x c, and 1 x d)	2 (1 x a and 1 x b)
Total reps	22	18

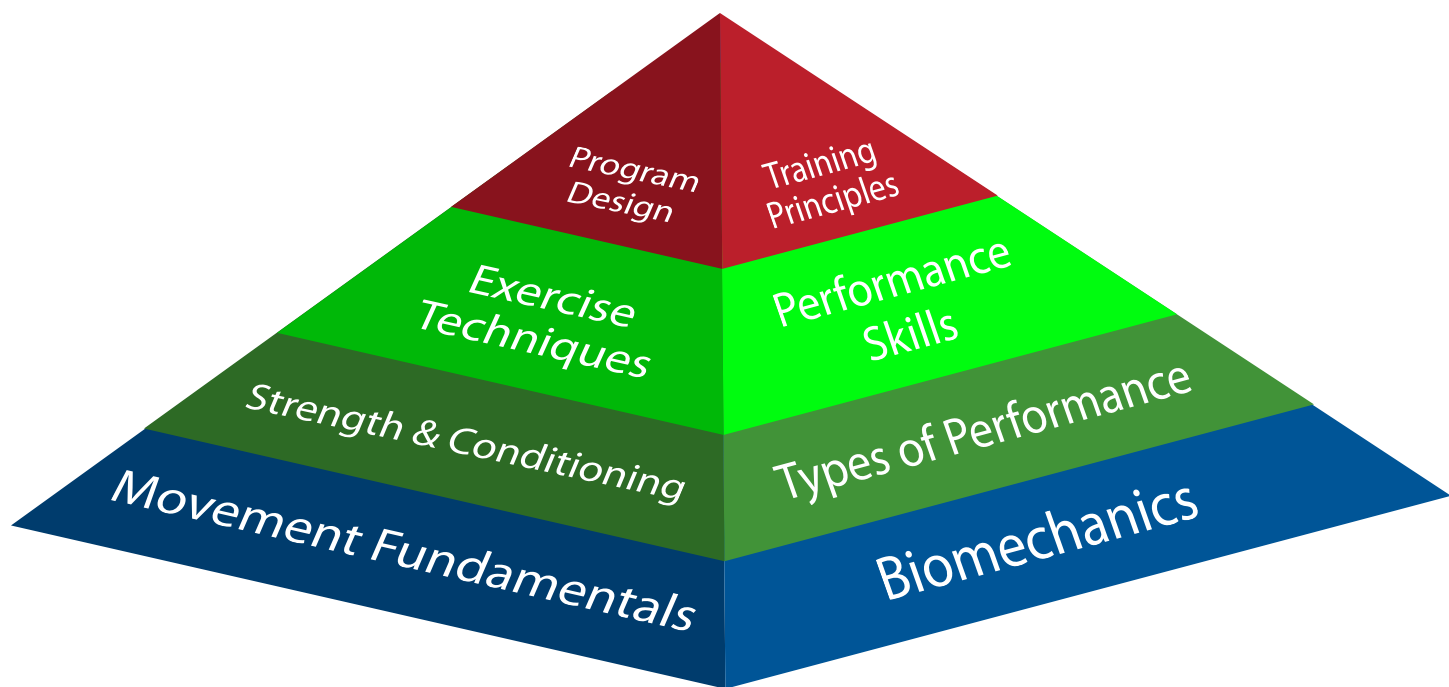
Week 8		
Drill	Day 1 Reps (HEAVY)	Day 2 Reps (LIGHT)
Rope or Ladder Routine (a, b, and c)	1 each	1 each
Bag Routine (a, b, and c)	1 each	1 each
Line Jump Routine (a, b, c, and d)	1 each	1 each
2 minute water break		
Pro-Agility	4 (2 left, 2 right)	3 (2 left, 1 right)
Three-Cone Drill	6 (3 left, 3 right)	4 (2 left, 2 right)
2 minute water break		
Sprint Ladder	6 (2 x a, 2 x b, 1 x d, and 1 x e)	4 (1 x a, 1 x b, 1 x d, and 1 x e)
Total reps	26	21

Week 9		
Drill	Day 1 Reps (HEAVY)	Day 2 Reps (LIGHT)
Rope or Ladder Routine (a, b, and c)	1 each	1 each
Bag Routine (a, b, and c)	1 each	1 each
Line Jump Routine (a, b, c, and d)	1 each	1 each
2 minute water break		
Pro-Agility	4 (2 left, 2 right)	2 (1 left, 1 right)
Nebraska Agility	4	2
Three-Cone Drill	4 (2 left, 2 right)	4 (2 left, 2 right)
2 minute water break		
Sprint Ladder	2 (1 x d and 1 x e)	2 (a and b)
Shuffle Ladder	2 (1 x a and 1 x b)	2 (1 x c and 1 x d)
Backpedal Ladder	2 (1 x a and 1 x b)	2 (1 x d and 1 x e)
Total reps	28	24

Week 10		
Drill	Day 1 Reps (HEAVY)	Day 2 Reps (LIGHT)
Rope or Ladder Routine (a, b, and c)	1 each	1 each
Bag Routine (a, b, and c)	1 each	1 each
Line Jump Routine (a, b, c, and d)	1 each	1 each
2 minute water break		
Nebraska Agility	4	2
Three-Cone Drill	6 (3 left, 3 right)	4 (2 left, 2 right)
2 minute water break		
Sprint Ladder	5 (1 x a, 1 x b, 1 x c, 1 x d, and 1 x e)	2 (1 x d and 1 x e)
Shuffle Ladder	3 (1 x a, 1 x d, and 1 x e)	4 (2 x a and 2 x b)
Backpedal Ladder	3 (1 x a, 1 x d, and 1 x e)	4 (2 x a and 2 x b)
Total reps	31	26

Week 11		
Drill	Day 1 Reps (HEAVY)	Day 2 Reps (LIGHT)
Rope or Ladder Routine (a, b, and c)	1 each	1 each
Bag Routine (a, b, and c)	1 each	1 each
Line Jump Routine (a, b, c, and d)	1 each	1 each
2 minute water break		
Nebraska Agility	4	2
Three-Cone Drill	6 (3 left, 3 right)	4 (2 left, 2 right)
2 minute water break		
Four-Corner Drill	6 (3 left, 3 right)	4 (2 left and 2 right)
Sprint Ladder	2 (1 x a and 1 x b)	3 (1 x a, 1 x c, and 1 x d)
Backpedal Ladder	2 (1 x d and 1 x e)	3 (1 x a, 1 x b, and 1 x c)
Total reps	30	26

Week 12		
Drill	Day 1 Reps (HEAVY)	Day 2 Reps (LIGHT)
Rope or Ladder Routine (a, b, and c)	1 each	1 each
Bag Routine (a, b, and c)	1 each	1 each
Line Jump Routine (a, b, c, and d)	1 each	1 each
2 minute water break		
Pro-Agility	4 (2 left, 2 right)	3 (1 left, 2 right)
Three-Cone Drill	4 (2 left, 2 right)	4 (2 left, 2 right)
Four-Corner Drill	6 (3 left, 3 right)	4 (2 left, 2 right)
2 minute water break		
Sprint Ladder	5 (1 x a, b, c, d, and e)	4 (1 x a, 1 x b, 1 x d, and 1 x e)
Backpedal Ladder	5 (2 x a, 2 x d, and 1 x c)	3 (1 x a, 1 x b, and 1 x c)
Total reps	34	28



Performance Pyramid